COURSE PROJECT:

Midterm Theory Report

Course: Introduction to Information Security
Instructor: Huynh Ngoc Tu, Ph.D

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1. Project Metadata

- **ClassID:** **Group ID:**
- **Project Title:** [e.g., "SQL Injection: Mechanics and Prevention"]
- **Student Name:** _____ **Student Name:** _____
- **Student ID:** _____ **Student ID:** _____
- **Topic Domain:** [e.g., Application Security / Network Security]

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2. Executive Summary (Abstract)

(Max 200 words)

> **Instruction:** Briefly describe the security concept or vulnerability you are researching. Summarize what it is, why it is dangerous (or useful), and what you plan to demonstrate in the final semester project.
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> *Example: "This report analyzes SQL Injection, a code injection technique used to attack data-driven applications. It occurs when malicious SQL statements are inserted into entry fields for execution. This project will explore the theoretical failure of input validation and culminate in a demonstration of a vulnerable login page."*

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3. Technical Background Concepts

(Min 300 words)

> **Instruction:** Before explaining the attack/defense itself, explain the **underlying technology** that makes it possible. Do not talk about the hack yet; talk about the system.
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> - *If your topic is Buffer Overflow:* Explain how computer memory (Stack and Heap) works. What is a pointer?
> - *If your topic is Phishing:* Explain email protocols (SMTP), headers, and how domain reputation works.
> - *If your topic is Encryption:* Explain the algorithm (encryption, decryption…) and the mathematical difference between Symmetric and Asymmetric keys.

4. System Architecture & Logic Diagram

Instruction: Insert a diagram here that visualizes the flow of data in your scenario. This can be a flow chart, a sequence diagram, or a network topology map. **You must create this diagram yourself or heavily annotate an existing one.**

- **Required:** Label the "Attacker," the "Victim/Server," and the specific "Vulnerability Point."

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5. The Mechanics of the Vulnerability (The "Why")

(Min 400 words)

Instruction: This is the core of your midterm report. Explain **technically** how the security failure happens.

- **Step-by-Step Analysis:** Walk through the logic. (e.g., "Step 1: User inputs data. Step 2: Server fails to sanitize. Step 3: Database interprets input as command.")
- **Code/Logic Snippet:** Provide a *theoretical* example of vulnerable code (pseudo-code is acceptable) or a configuration error.
- **The "root cause":** Is this a design flaw, an implementation bug, or human error?

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6. Impact Analysis (The CIA Triad)

Instruction: Analyze how this specific topic affects the three pillars of information security. Be specific.

- **Confidentiality:** [Does this allow unauthorized data access? e.g., Stealing passwords.]
- **Integrity:** [Does this allow data modification? e.g., Changing a bank balance.]
- **Availability:** [Does this crash the system? e.g., Denial of Service.]

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7. Proposed Methods /or Prevention & Mitigation

(Min 200 words)

Instruction: How do industry professionals fix or prevent this? Do not just say "use a firewall." Be technical.

- *Example:* "To prevent SQL Injection, developers must use **Prepared Statements** (Parameterized Queries) rather than string concatenation. This ensures the database treats user input strictly as data, not executable code."

8. Plan for Final Demonstration (*_required_)

Instruction: Briefly outline your plan for the end of the semester.

- **Tools required:** [e.g., Python, VirtualBox, Wireshark, Kali Linux]
- **Scenario:** [e.g., "I will set up a virtual machine acting as a bank server and attack it from a second VM."]
- **Success Criteria:** [e.g., "The demo is successful if I can extract the admin password hash."]

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9. References

Instruction: List at least **3 reputable sources**. (Wikipedia is not a primary source).

- [Format: IEEE or APA style]
- [e.g., NIST Publications, OWASP Documentation, RFC Standards, Academic Papers]